An AI-Driven Integrated Water Intelligence Framework For Sustainable And Equitable Water Systems

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Abstract*—Rural water supply systems face significant challenges due to declining groundwater availability, deteriorating water quality, distribution losses caused by pipeline leakages, and the absence of integrated decision-support mechanisms. Existing approaches typically assess groundwater resources, water quality, and infrastructure performance independently, resulting in fragmented monitoring and reactive management practices. This paper proposes an AI-driven Integrated Water Intelligence Framework for predictive and sustainable water management. The framework combines four complementary intelligence layers: Sustainability Intelligence, Risk Intelligence, Operational Intelligence, and Social Intelligence. Sustainability Intelligence evaluates groundwater vulnerability through a Groundwater Stress Index derived from recharge, extraction, availability, and future demand indicators. Risk Intelligence utilizes water quality parameters and environmental factors to assess contamination risks and compute a Water Quality Score. Operational Intelligence incorporates a YOLO-based leakage detection module to identify infrastructure-related water losses and support preventive maintenance planning. The outputs of these modules are integrated into an Integrated Water Risk Index (IWRI), which is subsequently modeled using a Graph Attention Network (GAT) to capture spatial dependencies among districts and generate explainable district-level risk predictions. The proposed framework enables holistic assessment of water security by integrating resource sustainability, water quality, infrastructure performance, and spatial interactions. The results demonstrate the potential of AI-enabled predictive governance for improving water resource planning, prioritizing interventions, and supporting sustainable rural water management.

Keywords— Graph Attention Network, Groundwater Stress Assessment, Integrated Water Risk Index, Predictive Water Governance, Sustainable Water Management, Water Quality Analysis, Water Leakage Detection.

I. Introduction

Groundwater is one of the most critical freshwater resources supporting agricultural, industrial production, and domestic water supply across India. With increasing population growth, rapid urbanization, agricultural intensification, and climate variability, groundwater resources are experiencing unprecedented stress. The damage is rapidly leading to the irreversible depletion of Earth's underground aquifers. [1]

India is currently the largest extractor of underground water globally, accounting for a significant share of worldwide groundwater withdrawals. In several regions, excessive extraction has led to declining water tables, deteriorating water quality, and growing concerns about long-term water security. Uttar Pradesh, India's most populous state, faces considerable groundwater management challenges due to extensive agricultural dependence, high population density, and uneven distribution of water resources. While many districts continue to rely heavily on groundwater for irrigation and domestic consumption, rising extraction rates, coupled with changing rainfall patterns, have led to localized groundwater stress. Here, the issue of over-extraction, along with rapid environmental changes and human activities, is also contributing to the deterioration of water quality, including elevated total dissolved solids (TDS), fluoride and sulfate contamination, and turbidity, further threatening the sustainability of groundwater resources [3].

Further, a significant portion of treated water is lost before reaching consumers due to pipeline leaks, aging infrastructure, and unauthorized connections. Water losses reduce the effective availability of already stressed groundwater resources. Leakage increases operational costs, energy consumption, and maintenance requirements. Existing leakage detection methods are often reactive rather than predictive. Most leakage monitoring systems operate independently from groundwater management frameworks

The existing method uses fragmented analysis of water, which separates the assessment of groundwater quality, quantity, and leakage as a separate problem. Most of the studies focus on recharge and extraction patterns, ultimately lacking integration, preventing a holistic understanding of regional water risk [4], [6].

Traditional groundwater risk assessment approaches primarily employ statistical methods or conventional machine learning models such as Random Forest, Support Vector Machines, and Gradient Boosting algorithms. Although these methods have demonstrated effectiveness in identifying groundwater-related patterns, they typically treat each administrative region as an independent observation. Such an assumption overlooks the inherently interconnected nature of the groundwater system, where the hydrological processes, aquifer characteristics, river basin interactions, and anthropogenic activities often influence neighboring regions. Consequently, models that ignore spatial dependencies may fail to capture important environmental relationships affecting groundwater sustainability.

Recent advances in Graph Neural Network (GNNs) provide an opportunity to model complex spatial interactions within environmental systems. By representing districts as nodes and their geographical or hydrogeological relationships as graph edges, GNNs can learn both local and neighboring influences simultaneously. Among these architectures, Graph

Attention Networks (GATs) [9], offer additional advantage through an attention mechanism that quantifies the relative importance of neighboring regions, thereby improving both predictive performance and interpretability.

The main contributions of this paper are:

1) Development of a unified framework integrating groundwater quantity, groundwater quality, and water leakage assessment.

2) Construction of a district-level graph representation for Uttar Pradesh.

3) Application of Graph Attention Networks for spatial risk prediction.

4) Development of a novel Integrated Water Risk Index (IWRI).

II. RELATED WORK

A. *Groundwater Stress Assessment*

Groundwater stress assessment has been widely studied using hydrogeological, statistical, and machine learning approaches. Traditional methods rely on groundwater recharge, annual extraction, and groundwater development indicators to evaluate resource sustainability. Several studies have employed machine learning algorithms such as Random Forest, Support Vector Machines, Artificial Neural Networks, and XGBoost to predict groundwater levels and identify regions vulnerable to overexploitation [7], [8]. While these approaches have demonstrated satisfactory predictive performance, they generally treat geographical regions as independent entities and fail to account for interactions among neighboring districts. Consequently, the influence of regional hydrological connectivity on groundwater availability remains insufficiently captured.

B. *Water Quality Assessment and Risk Analysis*

Water quality assessment is commonly performed using Water Quality Index (WQI) frameworks that are integrated with multiple physicochemical parameters such as pH, total dissolved solids (TDS), sulfate concentration, turbidity, hardness, nitrate concentration, and fluoride levels. Recently, studies have incorporated machine learning techniques for water quality classification and contamination prediction. Although these methods provide insights into groundwater suitability, they primarily focus on quality indicators without considering groundwater availability, extraction patterns, or infrastructure-related losses. As a result, they offer only a partial representation of overall water resource risk.

C. *Water Leakage Detection and Distribution Loss Analysis*

Water distribution losses due to pipeline leakages represent a major challenge for urban and rural water management systems. Traditional leakage detection techniques rely on `pressure monitoring, acoustic sensing, and flow analysis. More recently, computer vision and deep learning approaches, particularly YOLO-based object detection models, have demonstrated promising results in identifying visible leakages and pipeline defects. Despite significant advancements in leakage detection accuracy, existing studies typically operate as standalone infrastructure monitoring systems and remain disconnected from broader groundwater sustainability and water resource management frameworks [12] ,[13], [14], [15], [16].

D. *Graph Neural Network in Environmental Monitoring*

Graph Neural Networks (GNNs) have emerged as powerful tools for modelling complex spatial dependencies in interconnected systems. Applications of GNNs have been reported in transportation forecasting, climate analysis, environmental monitoring, and resource management. Among various GNN architectures, Graph Attention Networks (GATs) have gained popularity due to their ability to learn adaptive neighbor importance through attention mechanisms. Several studies have demonstrated the effectiveness of GATs in capturing spatial correlations while providing interpretable explanations. However, the application of Graph Attention Networks for integrated groundwater risk assessment remains limited, particularly in the context of combining groundwater quantity, water quality, and infrastructure-related distribution losses within a unified framework [9], [10], [11].

E. *Research Gap*

A review of existing literature reveals three significant limitations. First, groundwater quantity assessment, water quality evaluation, and leakage detection are predominantly studied as independent problems. Second, most predictive models ignore spatial dependencies between neighbouring regions despite the interconnected nature of groundwater systems. Third, existing frameworks often lack explainability, limiting their usefulness for policy formulation and resource planning.

To address this, the work proposes an Explainable Graph Attention Network-based Integrated Water Risk Assessment Framework that combines groundwater stress indicators, water quality parameters, and leakage impact factors into a unified district-level spatial learning architecture. The proposed framework enables holistic risk assessment while simultaneously capturing regional dependencies and providing interpretable predictions for sustainable water resource management.

III. STUDY AREA AND DATASETS

A. *Study Area*

Uttar Pradesh, located in northern India, was selected as the study area due to dependence on groundwater resources and increasing concerns regarding groundwater sustainability. The state covers an area of approximately 240,928 km^2 and consists of 75 administrative districts [2], [23].

Groundwater serves as a primary source of irrigation, domestic water supply, and industrial consumption across the state. The hydrogeological conditions of Uttar Pradesh are strongly influenced by Indo-Gangetic alluvial plains, which contains extensive aquifer system supporting agricultural activities. However, rapid population growth, intensive agricultural practices, urbanization, and climate variability have resulted in increasing pressure on groundwater resources.

Several districts have reported declining groundwater levels, excessive extraction rates, and deteriorating water quality conditions. The study includes districts distributed across different hydrogeological zones, allowing the proposed framework to evaluate diverse groundwater conditions and spatial interactions across the state.

Fig. 1. District map of Uttar Pradesh showing the study area.

Uttar Pradesh was selected as the study area due to its extensive dependence on groundwater resources and its diverse hydrogeological conditions. The key characteristics of the study area are summarized in Table I.

TABLE I.

STUDY AREA CHARACTERISTICS

Parameters	***Value***	***Units***
Area	240,928	km^2
Districts	75	-
Population	240,000,000	Persons
Groundwater Dependency	>70	%
Major River Basins	Ganga, Yamuna	-

Source: Survey of India

As shown in Table I, Uttar Pradesh is one of the largest and most populous states in India. The presence of extensive agricultural activities and a large population result in significant dependence on groundwater resources for irrigation, domestic consumption, and industrial applications. These characteristics make Uttar Pradesh an appropriate region for evaluating groundwater sustainability and water risk assessment frameworks.

TABLE II.

GROUNDWATER DEPENDENCY ACROSS MAJOR SECTOR IN UTTAR PRADESH

Sector	***Share (%)***
Agriculture/Irrigation	70
Urban Domestic Water Supply	75-80*
Rural Domestic Supply	90-95*
Industrial Water Demand	95

*Percentage of drinking water demand met through groundwater

Source: CGWB and NITI Aayog Reports.

As shown in Table II, groundwater serves as the primary source of water for agricultural activities and drinking water supply across Uttar Pradesh. Rural regions exhibit particularly high dependence on groundwater resources, emphasizing the importance of sustainable groundwater management strategies for ensuring long-term water security.

B. Groundwater Resource Dataset

The groundwater resource information was obtained from district-wise groundwater assessment reports published by the Central Groundwater Board (CGWB). These reports provide standardized estimates of groundwater availability, extraction, recharge, and sustainability indicators. The groundwater dataset represents the quantitative status of groundwater resources and serves as the primary input for groundwater stress estimation [2], [22].

TABLE III.

GROUNDWATER RESOURCE FEATURE

Features	***Unit***
Recharge from Rainfall during Monsoon	Ham
Recharge from other sources during Monsoon	Ham
Recharge from Rainfall During Non-Monsoon Season	Ham
Recharge from Other Sources During Non-Monsoon Season	Ham
Annual Replenishable Groundwater Resources	Ham
Annual Natural Discharge	Ham
Net Annual Groundwater Availability	Ham
Draft Due to Irrigation Needs	Ham
Draft Due to Domestic and Industrial Water Supply Needs	Ham
Total Annual Draft	Ham
Projected Demand for Domestic and Industrial uses upto 2025	Ham
Groundwater Availability for Future Irrigation Use	Ham
Stage of Groundwater Development	%

Source: Central Ground Water Board (CGWB)

Recharge-related variables quantify groundwater replenishment, whereas draft-related variables represent groundwater extraction pressure. These indicators form the basis for the groundwater stress assessment module.

Fig. 2. Groundwater resource assessment workflow.

Figure 2 illustrates the transformation of raw groundwater resource variables into groundwater stress indicators. Recharge-related parameters contribute to groundwater availability estimation, while extraction-related variables are used to quantify groundwater utilization pressure.

C. *Water Quality Dataset*

The water quality dataset consists of physicochemical parameters commonly used for evaluating groundwater suitability and contamination risk. Water quality parameters were obtained from the publicly available Water Potability Dataset [21]. These parameters were selected because they represent important indicators of drinking water quality and have been widely utilized in groundwater assessment studies. The dataset features are summarized in Table IV.

TABLE IV.

WATER QUALITY DATASET FEATURES

Parameters	***Units***	***Description***
pH	-	Acidity or alkalinity of water
Hardness	mg/L	Concentration of dissolved calcium and magnesium salts
Solids (TDS)	mg/L	Total dissolved solids present in water
Chloramines	mg/L	Disinfectant compounds used in water treatment
Sulfate	mg/L	Sulfate ion concentration
Conductivity	µS/cm	Electrical conductivity of water
Organic Carbon	mg/L	Amount of dissolved organic carbon
Trihalomethanes	µg/L	By-products formed during chlorination
Turbidity	NTU	Measure of suspended particles
Potability	Binary	Drinking water suitability

D. *Leakage Detection Dataset*

Water distribution losses caused by pipeline leakages represent a significant challenge in achieving sustainable water resource management. In many regions, substantial quantities of treated water are lost due to undetected leakages within distribution networks, resulting in reduced water availability and increased operational costs. To incorporate infrastructure-related losses into the proposed Integrated Water Risk Assessment Framework, a computer vision-based leakage detection module was developed using the Water Pipes Dataset.

The Water Pipes Dataset consists of annotated images depicting water pipeline leakages captured under diverse environmental conditions. The leakage detection module utilized the Water Pipes Dataset consisting of annotated pipe images for leakage detection tasks [20]. The dataset includes images obtained from real-world pipeline systems, web-based image repositories, and simulated environments. To improve model robustness and enhance generalization performance, multiple augmentation techniques were applied to the original dataset. The resulting dataset characteristics are summarized in Table VI.

TABLE V.

LEAKAGE DETECTION DATASET CHARACTERISTIC

Attributes	***Value***
Dataset Name	Water Pipes Dataset
Source	Kaggle
Dataset Creator	Tareq Alhmiedat et al.
Dataset Type	Image Dataset
Original Images	197
Argumented Images	764
Image Format	JPG
Annotation Format	YOLO
Task	Water Leakage Detection
Image Sources	Real-world photographs, Google Images, and simulation environment
Number of Classes	1(Leakage)
Argumentation Techniques	Rotation, brightness adjustment, contrast adjustment, cropping/zooming, noise addition, background color alteration, grayscale conversion

Source: Water pipes Dataset (Kaggle)

As shown in Table V, the dataset contains both original and augmented images, enabling the model to learn leakage patterns under varying lighting conditions, viewpoints, and background complexities. The use of augmentation techniques increases dataset diversity and reduces the likelihood of overfitting during model training.

To further improve the robustness of the leakage detection model, several image transformation and augmentation strategies were employed. These techniques simulate real-world operational conditions and improve the ability of the model to detect leakages in unseen scenarios. The augmentation techniques utilized in this study are summarized in Table VI.

TABLE VI.

DATA AUGMENTATION TECHNIQUES

Techniques	*Purpose*
Rotation	Simulate different camera angles
Brightness Adjustment	Handle illumination variations
Contrast Adjustment	Improve robustness to lighting changes
Cropping/ Zooming	Improve object localization
Noise Addition	Enhance model generalization
Background Color Alteration	Reduce background bias
Grayscale Conversion	Improve robustness to color variations
Image Sources	Real-world photographs, Google Images, and simulation environment
Number of Classes	1(Leakage)

Source: Water pipes Dataset (Kaggle)

The application of these augmentation methods enhances the variability of the training data and enables the model to learn discriminative leakage features across different environmental settings. Such variability is essential for ensuring reliable deployment of the proposed framework in practical water distribution systems.

Figure 3 presents representative examples from the Water Pipes Dataset along with their corresponding annotations. The images demonstrate the diversity of leakage appearances, including differences in leakage size, intensity, background conditions, and viewing perspectives.

Fig. 3. Sample images from the Water Pipes Dataset with leakage annotations.

The leakage detection outputs generated by the trained YOLO-based model were subsequently transformed into district-level leakage risk indicators and integrated with groundwater stress and water quality parameters during the construction of the proposed Integrated Water Risk Index (IWRI). This integration enables the framework to simultaneously account for groundwater sustainability, water quality degradation, and infrastructure-related water losses within a unified risk assessment model.

E. Rainfall and Demographic Data

Rainfall and demographic factors play a crucial role in influencing groundwater recharge, extraction patterns, and long-term sustainability. Rainfall directly affects groundwater replenishment through infiltration and percolation processes, whereas population-related indicators influence groundwater demand for domestic, agricultural, and industrial activities. Therefore, climatic and demographic variables were incorporated into the proposed framework to capture both supply-side and demand-side factors affecting groundwater resources. District-wise rainfall statistics were obtained from the India Meteorological Department (IMD) [18], [24].

District-wise rainfall data were collected from meteorological sources and integrated with groundwater resource information to represent recharge potential across Uttar Pradesh. Similarly, demographic indicators including total population and population density were utilized to characterize regional water demand and anthropogenic pressure on groundwater systems. The inclusion of these variables enables the proposed model to better capture spatial variations in groundwater stress and sustainability.

The climatic and demographic variables used in this study are summarized in Table VII

TABLE VII.

RAINFALL AND DEMOGRAPHIC FEATURES

Parameters	*Units*	*Description*
Annual Rainfall	mm	Average annual rainfall received by a district
Population	Persons	Total district population
Population Density	Persons/km^2	Population concentration within a district

As shown in Table VII, rainfall represents the primary natural source of groundwater recharge, while population-related variables act as indicators of groundwater demand. The integration of these factors provides a more comprehensive representation of regional groundwater dynamics and supports the development of a robust district-level risk assessment framework. Population and demographic indicators were obtained from Census of India records [23].

F. Dataset preprocessing

The collected datasets originated from multiple sources and exhibited differences in structure, scale, and data quality. Consequently, a comprehensive preprocessing pipeline was implemented to ensure consistency, reliability, and compatibility among all datasets prior to model development. The preprocessing stage included missing value treatment, feature standardization, district-wise aggregation, and feature engineering.

Missing values present within groundwater and water quality datasets were handled using median imputation to minimize the influence of outliers while preserving the underlying data distribution. Following imputation, all numerical features were normalized using Min-Max scaling to ensure comparable feature ranges and improve model convergence during training.

Since the proposed framework operates at the district level, data from multiple sources were aggregated using district identifiers. Groundwater resource indicators, water quality parameters, rainfall information, demographic statistics, and leakage detection outputs were merged to create a unified district-wise dataset. This integration enabled the construction of node-level feature representations required for graph-based learning.

Furthermore, several derived indicators were generated through feature engineering to better characterize groundwater sustainability and water-related risks. These engineered features were subsequently utilized during Integrated Water Risk Index computation and graph construction.

TABLE VIII.

DATA PREPROCESSING OPERATIONS

OPERATION	***Purpose***
Missing Value Imputation	Handle incomplete records
Min-Max Normalization	Standardize feature ranges
District-wise Aggregation	Create district-level observations
Dataset Integration	Merge multiple datasets
Feature Engineering	Generate risk-related indicators

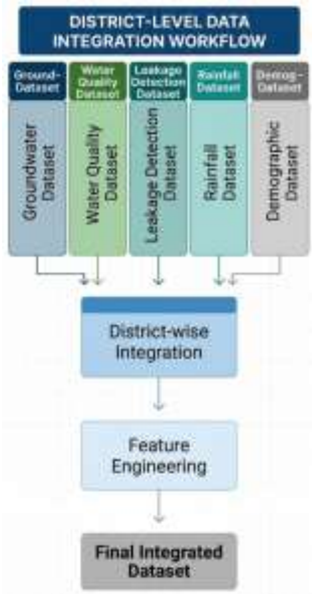

Fig. 4. Dataset integration and preprocessing workflow.

The preprocessing pipeline transformed heterogeneous datasets into a unified district-level representation suitable for machine learning and graph-based analysis. The resulting dataset served as the foundation for the computation of groundwater stress indicators, water quality scores, leakage risk metrics, and the subsequent construction of the Graph Attention Network-based Integrated Water Risk Assessment Framework.

IV. PROPOSED METHODOLOGY

The proposed framework integrates groundwater resource assessment, water quality evaluation, leakage detection, climatic factors, and demographic indicators within a unified spatial learning architecture. The overall methodology consists of six major stages: groundwater stress assessment, water quality assessment, leakage detection, Integrated Water Risk Index computation, graph construction, and district-level risk prediction using a Graph Attention Network (GAT). The workflow of the proposed framework is illustrated in Fig. 5.

A. System Architecture

Fig. 5. Overall architecture of the proposed framework.

B. Groundwater Stress Assessment

Groundwater stress represents the imbalance between groundwater availability and groundwater extraction. To quantify district-level groundwater sustainability, multiple stress indicators were derived from the groundwater resource dataset.

1) Groundwater Stress Index

$$GSI = \frac{Total\ Annual\ Draft}{Net\ Annual\ Groundwater\ Availability} \quad (1)$$

Where:

- Total Annual Draft = total groundwater extraction
- Net Annual Groundwater Availability = usable groundwater resources

2) Future Stress Index (FSI)

$$FSI = \frac{Projected\ Demand}{Net\ Annual\ Groundwater\ Availability} \quad (2)$$

This indicator estimates future pressure on groundwater resources.

3) Irrigation Dependency Index (IDI)

$$FSI = \frac{Draft\ Due\ To\ Irrigation}{Total\ Annual\ Draft} \quad (3)$$

This metric measures the proportion of groundwater utilized for agricultural activities.

4) Groundwater Stress Score

$$GSS = \frac{GSI+FSI+IDI}{3} \quad (4)$$

The resulting score was normalized to a range of 0-100

C. Water Quality Assessment

Groundwater quality plays a critical role in determining the suitability of water for drinking, agricultural, and industrial applications. Deterioration in water quality can adversely affect public health and ecosystem sustainability. Therefore, a Water Quality Assessment Module was developed to quantify district-level groundwater quality risk using physicochemical parameters obtained from the water quality dataset.

The selected parameters include pH, hardness, total dissolved solids (TDS), chloramines, sulfate, conductivity, organic carbon, trihalomethanes, and turbidity. These indicators collectively represent the chemical and physical characteristics of groundwater and are widely used for water quality evaluation.

1) Feature Normalization

Since the water quality parameters possess different units and numerical ranges, Min-Max normalization was applied to ensure comparability among features.

$$N_i = \frac{X_i - X_{min}}{X_{max} - X_{min}} \quad (5)$$

where:

- N_i= normalized value
- X_i= observed parameter value
- X_{min}= minimum observed value
- X_{max}= maximum observed value

The normalization process transforms all parameters into a common range of 0 to 1.

2) Water Quality Score Computation

The normalized parameters were aggregated to compute a district-level Water Quality Score (WQS).
Using weighted scoring

$$\text{WQS} = \sum_{i=1}^{n} w_i\ N_i \quad (6)$$

subject to

$$\sum_{i=1}^{n} w_i = 1 \quad (7)$$

where:

- w_i= weight assigned to parameter i
- N_i= normalized value

This method accounts for the varying significance of different water quality indicators.

3) Parameter Weight Assignment

TABLE IX.

WEIGHT ASSIGNED TO WATER QUALITY PARAMETER

Parameters	***Weights***
pH	0.10
Hardness	0.10
Solids (TDS)	0.15
Chloramines	0.10
Sulfate	0.15
Conductivity	0.10
Organic Carbon	0.10
Trihalomethanes	0.10
Turbidity	0.10

4) Water Quality Risk Classification

TABLE X.

WEIGHT QUALITY RISK CATEGORY

WQS Range	***Category***
0-25	Good
25-50	Moderate
50-75	Poor
75-100	Critical

The computed Water Quality Score provides a quantitative measure of groundwater quality conditions at the district level. Higher WQS values indicate greater water quality degradation and increased contamination risk. The resulting score is subsequently integrated with groundwater stress indicators and leakage detection outputs during the computation of the Integrated Water Risk Index (IWRI).

Fig. 6. Water quality assessment workflow.

D. *Leakage Detection Module*

Water leakages in distribution networks result in substantial water losses and reduce the overall efficiency of water management systems. To quantify infrastructure-related losses, a deep learning-based object detection framework was employed to automatically identify visible pipeline leakages from image data. The leakage detection outputs were subsequently integrated into the proposed Integrated Water Risk Index (IWRI).

1) YOLO-Based Detection Framework

The leakage detection task was formulated as an object detection problem using the YOLO (You Only Look Once) architecture. YOLO performs object localization and classification in a single forward pass, making it suitable for real-time leakage monitoring applications. The leakage detection framework was implemented using the YOLO object detection architecture due to its high accuracy and real-time detection capabilities [15], [16].

2) Training Procedure

The leakage dataset was divided into training, validation, and testing subsets. During training, image augmentation techniques including rotation, brightness adjustment, contrast enhancement, and noise injection were applied to improve model robustness.

TABLE XI.

YOLO TRAINING PARAMETERS

Parameters	***Value***
Model	YOLOv8n
Epochs	100
Batch Size	16
Image Size	640*640
Optimizer	AdamW
Learning Rate	0.001
Confidence Threshold	0.25

3) Leakage Score Compuatation

The detected leakages were transformed into a district-level leakage indicator.

$$LS = \frac{N_L}{N_{max}} \quad (8)$$

where:

- LS= Leakage Score
- N_L= Number of detected leakages
- N_{max}= Maximum observed leakages

The score was normalized between 0 and 1.

4) Evaluation Metrics

Model performance was evaluated using standard object detection metrics.

$$Precision = \frac{TP}{TP + FP} \quad (9)$$

$$Recall = \frac{TP}{TP + FN} \quad (10)$$

$$F1 = \frac{2(Precision)(Recall)}{Precision + Recall} \quad (11)$$

where:

- TP = True Positives
- FP = False Positives
- FN = False Negatives

Additionally, Mean Average Precision (mAP) was used to evaluate detection accuracy.

The resulting leakage scores provide a quantitative estimate of infrastructure-related water losses and serve as an important component of the Integrated Water Risk Index. These scores were subsequently combined with groundwater stress and water quality indicators to enable holistic water risk assessment.

E. *Integrated Water Risk Index*

Existing water assessment approaches typically evaluate groundwater availability, water quality, and infrastructure losses independently. However, sustainable water management requires a unified assessment framework capable of simultaneously considering these interdependent factors. To address this limitation, an Integrated Water Risk Index (IWRI) was developed.

Components of IWRI

The proposed index combines:

Groundwater Stress Score (GSS)

Derived from:

- Groundwater Stress Index
- Future Stress Index
- Irrigation Dependency Index

Water Quality Score (WQS)

Derived from:

- pH
- Hardness
- Sulfate

- Turbidity
- Other physicochemical parameters

Leakage Score (LS)
Derived from:

- YOLO-based leakage detection outputs

IWRI Formulation

$$IWRI = 0.4(GSS) + 0.4(WQS) + 0.2(LS) \quad (12)$$

where:

- GSS = Groundwater Stress Score
- WQS = Water Quality Score
- LS = Leakage Score

The weights were selected to emphasize groundwater sustainability and water quality while still accounting for infrastructure-related losses.

F. Graph Construction

Groundwater systems exhibit strong spatial dependencies, where groundwater conditions in one district can influence neighboring districts. To capture these interactions, a graph-based representation of Uttar Pradesh was constructed. Each district was represented as a node in the graph, while spatial relationships between districts were represented as edges. Graph-based learning approaches enable the modeling of spatial dependencies that are often overlooked by traditional machine learning methods [9]–[11]. Node features included groundwater stress indicators, water quality scores, leakage scores, rainfall, population, and demographic characteristics
Node Feature Vector

$$X_i = [GSS, WQS, LS, Rainfall, Population, Density] \quad (13)$$

where X_i represents the feature vector of district i.

Edge Definition
Two districts were connected if the distance between their administrative centers was less than 100 km.

$$d_{ij} < 100 \quad (14)$$

where d_{ij} denotes the geographical distance between districts i and j.

Edge Weighting
Distance-based edge weights were assigned using:

$$w_{ij} = e^{-d_{ij}/100} \quad (15)$$

where w_{ij} represents the influence of neighboring district j on district i.
The resulting weighted graph captures spatial relationships among districts and serves as the input to the proposed Graph Attention Network for district-level water risk prediction.

G. Graph Attention Network (GAT)

To capture spatial dependencies among neighboring districts, a Graph Attention Network (GAT) was employed [11]. Unlike traditional graph convolution methods, GAT dynamically assigns attention weights to neighboring districts, enabling the model to learn the relative importance of different spatial interactions.

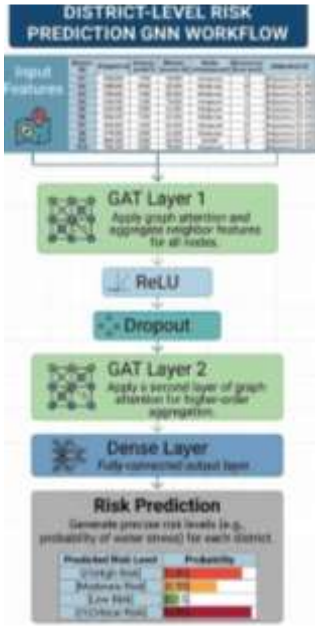

Fig. 7. Proposed graph Attention Network architecture

Attention Mechanism

$$\alpha_{ij} = \frac{\exp(e_{ij})}{\sum_{k \in N(i)} \exp(e_{ik})} \quad (16)$$

where:

- α_{ij} = attention coefficient
- $N(i)$ = neighboring districts
- e_{ij} = learned importance score

The final output of the network is the predicted district-level Integrated Water Risk Index.

V. EXPERIMENTAL SETUP

A. Hardware and Software Environment

The proposed framework was implemented in Python using PyTorch Geometric and trained on Google Colab with NVIDIA Tesla T4 GPU acceleration. Data preprocessing and analysis were performed using NumPy, Pandas, and Scikit-learn, while the leakage detection module was developed using YOLOv8.

TABLE XII.

HARDWARE AND SOFTWARE CONFIGURATION

Component	***Specification***
GPU	NVIDIA Tesla T4
RAM	16GB
Platform	Google Colab
Framework	PyTorch Geometric
Object Detection Model	YOLOv8
Language	Python

B. Training Configuration

The Graph Attention Network was trained using the Adam optimizer with a learning rate of 0.001. The model consisted of two GAT layers followed by a dense output layer.

TABLE XIII.

MODEL TRAINING PARAMETERS

Parameters	*Value*
Learning Rate	0.001
Optimizer	Adam
Epochs	100
Batch Size	16
Hidden Units	64
Dropout	0.2

C. Dataset Split

The integrated district-level dataset was divided into training, validation, and testing subsets using a 70:15:15 ratio.

TABLE XIV.

DATASET PARTITIONING

Dataset	*Percentage (%)*
Training	70
Validation	15
Testing	15

D. Evaluation Metrics

Model performance was evaluated using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and the Coefficient of Determination (R^2).

$$MAE = \frac{1}{N}\sum | y_i - \hat{y}_i | \qquad (17)$$

$$RMSE = \sqrt{\frac{1}{N}\sum(y_i - \hat{y}_i)^2} \qquad (18)$$

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2} \qquad (19)$$

For the leakage detection module, Precision, Recall, F1-Score, and mAP were used to assess object detection performance.

VI. RESULT AND DISCUSSION

A. Model Performance

The proposed Graph Attention Network (GAT)-based framework was evaluated using MAE, RMSE, and R^2score. The results demonstrate the effectiveness of integrating groundwater stress, water quality, and leakage information within a graph-based learning framework.

TABLE XV.

MODEL PERFORMANCE RESULT

Metric	*Value*
MAE	5.13
RMSE	7.26
R^2 Score	0.86

The obtained results indicate that the proposed model effectively captures both local district characteristics and spatial dependencies among neighboring districts, resulting in accurate district-level water risk prediction.

B. District-Level Risk Analysis

The Integrated Water Risk Index (IWRI) was computed for all districts and classified into risk categories. Districts exhibiting high groundwater extraction, poor water quality, and significant leakage losses were identified as high-risk regions.

TABLE XVI.

TOP HIGH-RISK DISTRICTS

District	*IWRI Score*	*Risk Category*
Lucknow	82	Critical
Kanpur	78	Critical
Prayagraj	71	Poor
Varanasi	62	Poor
Agra	45	Moderate

The results indicate that districts with high groundwater development and increasing demand tend to exhibit elevated risk levels.

C. Explainability Analysis

The attention mechanism of the Graph Attention Network was utilized to identify influential neighboring districts. The learned attention coefficients revealed that geographically connected districts significantly contribute to risk prediction.

For example, Kanpur exhibited the highest influence on the groundwater risk prediction of Lucknow, highlighting the importance of spatial interactions in regional water management.

D. Discusion

The results demonstrate that integrating groundwater stress, water quality degradation, and infrastructure-related water losses provides a more comprehensive assessment of water security compared to traditional single-factor approaches. Furthermore, the graph-based framework successfully captures spatial dependencies between neighboring districts, enabling more accurate and explainable risk predictions. These findings highlight the potential of the proposed framework for supporting district-level water resource planning and sustainable groundwater management.

VII. LIMITATIONS

Although the proposed work demonstrates the potential of integrating groundwater stress assessment, water quality evaluation, and leakage detection within a unified Graph Attention Network architecture, several limitations should be acknowledged.

A. Data availability and quality

The performance of the proposed framework is highly dependent on the availability, completeness, and reliability of the district-level dataset. Groundwater resources reports on water quality measurements; demographic information is often collected by different agencies at varying temporal and spatial resolutions. In some districts, missing observations and inconsistencies in reporting may introduce uncertainty into the risk assessment process.

B. District-level aggregate assumption

The proposed framework aggregates groundwater and water quality indicators at the district level to facilitate regional-scale analysis. While this approach improves computational efficiency and enables statewide assessment, it may obscure local variations within districts. Groundwater assessment can vary significantly across blocks and villages, and these fine-scale variations are not explicitly represented in the current study.

C. Graph construction assumption

Spatial relationships between districts are modelled using geographical proximity and distance-based connectivity. Although a distance-based graph provides an effective approximation of regional interactions, actual groundwater movement is influenced by hydrological characteristics and geological formations. Therefore, the constructed graph may not fully represent the complexity of real-world groundwater dynamics.

D. Leakage detection constraints

The model relies on the availability of annotated imagery and visible leakage patterns. Underground leakages or sensor-inaccessible faults may remain undetected, potentially affecting the accuracy of distribution loss estimation.

Despite these limitations, the proposed framework provides a scalable and interpretable foundation for integrated groundwater risk and decision support.

VIII. FUTURE WORK

A. Regional, multi-state, and national integration

District-level analysis provides valuable regional insights. With more data, the model can be scaled on the block level and further extended to neighbouring states and eventually develop a nationwide groundwater risk monitoring platform capable of supporting large-scale water governance initiatives.

B. Real-time IoT Integration

Future work will focus on integrating real-time data streams from Internet of Things (IoT) devices, including groundwater monitoring sensors, smart water meters, pressure sensors, and rainfall stations. Such integration would enable continuous monitoring and real-time risk assessment.

C. Dynamic Graph Learning.

The current structure employs a static graph structure based on geographical relationships. Future research may explore a dynamic graph neural network capable of adapting graph structure over time in response to changing environmental conditions, extraction patterns, and climatic variability.

D. Water Resource Level Modelling

While the district-level analysis provides valuable regional insights, groundwater systems are fundamentally governed by aquifer characteristics. Future extensions of this work will incorporate aquifer boundaries, geological formations, and subsurface connectivity to improve spatial representation and predictive accuracy.

IX. CONCLUSION

The paper presented an explainable Graph Attention Network-based framework for Integrated Water Risk Assessment in Uttar Pradesh. Unlike traditional approaches that independently analyse groundwater quality and quantity, and distribution losses, the proposed system combines these dimensions within a unified architecture.

The study introduces a novel Integrated Water Risk Index (IWRI) by incorporating groundwater stress indicators, water quality parameters, and leakage impact scores. Districts were represented as nodes within the graph structure, enabling the Graph Attention Network to capture interconnected dependencies and regional interactions that are mostly ignored by traditional machine learning methods.

Furthermore, the attention mechanism provides interpretable insights into neighbouring district influences, resulting in the enhancement of transparency and practical utility of the framework. The use of computer vision-based leakage detection expanded the scope of groundwater assessment by considering the infrastructure-related distribution losses alongside environmental factors.

The proposed framework offers a scalable and explainable decision-support system for policymakers, water resource managers, and environmental agencies. By enabling district-level risk prediction, regional influence analysis, and integrated water resource evaluation, the framework contributes toward more informed and sustainable groundwater management strategies.

The results demonstrate that spatially aware artificial intelligence techniques have significant potential to improve groundwater governance and support long-term water security planning in data-driven environmental management systems.

Acknowledgment

I would like to express my sincere gratitude to the Central Ground Water Board, Kaggle, IMD, Census of India, and other government agencies whose publicly available datasets contributed significantly to this study.
I would like to acknowledge the support of my parents, my respective academic institutions, and my faculty for their constant support and encouragement.
The computational resources Google Colab, Kaggle, and Python-based libraries pyTorch, YOLO and associated open-source libraries are also gratefully acknowledged.

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